

Dayananda Sagar College of Engineering

Department of Computer Science & Engineering

INTERNSHIP-SEC (22CS82) REVIEW – 1

2025-2026

Name: Jyothika Lakshmi M

USN: 1DS22CS097

**Name of the SEC Courses: Skill Enhancement with Data Structure and
Algorithm – (C language),
React Full Stack (Web/App development skills)**



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CONTENT

1. COURSE ENROLLMENT & COMPLETION
2. CERTIFICATION QUALITY
3. KNOWLEDGE & CONCEPT UNDERSTANDING
4. APPLICATION / MINI ASSIGNMENT / CASE STUDY



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COURSE ENROLLMENT & COMPLETION

Enrolled in two Skill Enhancement Courses:

Skill Enhancement with Data Structures and Algorithms (C Language)
React Full Stack Development (Web/Application Development)

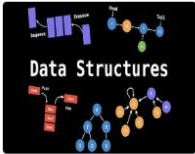
Successfully completed learning modules in both courses
Completed required assessments (MCQ-based) for each course
Currently eligible and preparing for final certification exam



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COURSE ENROLLMENT & COMPLETION

The screenshot displays the 'Center for Online Education' interface for VTU - Belagavi. The header includes the university logo, name, and a search bar. The main content area lists two courses with their progress status.

Course	Progress	Action
 Data Structures (Credits - 03) Skill enhancement with Data structure algorithm - (C language) ★ Give Review Apply for Exam	100.00% 	Completed
 React (Credits - 03) React Full stack (Web/App development Skills) ★ Give Review Apply for Exam	100.00% 	Completed

Additional UI elements include a 'Browse More Course' button, a 'Sort By' dropdown, and a Windows watermark at the bottom right.



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CERTIFICATION QUALITY

Certification will be awarded upon successful completion of final exam

- Courses designed with industry-relevant curriculum
- Focus on both theoretical and practical knowledge
- Assessment-based learning ensures concept clarity

Certification validates skills in:

Programming & Problem Solving
Full Stack Web Development



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KNOWLEDGE & CONCEPT UNDERSTANDING

Data Structures and Algorithms

- Developed strong foundation in **structured programming using C**, focusing on:
 - Memory management
 - Pointer-based logic
 - Efficient control flow
- Understood **core data structures from a problem-solving perspective**:
 - Arrays → Used for indexed data handling and optimization
 - Linked Lists → Dynamic memory usage & flexibility



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KNOWLEDGE & CONCEPT UNDERSTANDING

- Stack & Queue → Real-world use cases like recursion, scheduling
- Introduced to **algorithmic thinking**:
- Breaking problems into smaller steps
- Designing step-by-step solutions before coding
- Focus on **efficiency**:
- Basic understanding of **time & space complexity**
- Importance of optimizing code for large-scale data
- Improved logical thinking and coding skills



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KNOWLEDGE & CONCEPT UNDERSTANDING

React Full Stack

1. Frontend Development with React

- Gained hands-on understanding of **modern frontend architecture using React**
- Built reusable and modular UI using:
 - Functional components
 - Component-based design
- Learned **state-driven UI rendering**, which is core to modern apps:
 - State management
 - Props-based data flow
- Understood **event-driven programming** in web apps:
 - Handling user interactions
 - Dynamic updates without page reload



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2. Full Stack & Application Development Concepts

- **Understood client-server architecture:**
 - How frontend communicates with backend APIs
- Learned **data flow in applications:**
 - Request → Processing → Response
- Exposure to:
 - API integration
 - Data handling between layers

Learned how real-world web applications are structured
Improved ability to design simple web applications



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APPLICATION / MINI ASSIGNMENT / CASE STUDY

Practical Application

Completed MCQ-based assessments for both courses
Applied learned concepts to solve logical problems
Strengthened real-world problem-solving ability

Skills Acquired

Problem-solving & logical thinking
Frontend development using React
Basic full stack development knowledge
Understanding of real-world application development



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Courses

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Exam Administration Academics

(Credits - 03) Skill enhancement with Data structure algorithm - (C language)

33 lectures 9 sections 14 h 32 m 18 s

Quiz - 1

Total Score: 25 | Your Score: 22

1. Why are data structures important in computer science and programming?

- ☐ Only for organizing code
- ☐ Only for optimizing algorithms
- ☒ For organizing and managing data efficiently ✓
- ☐ For aesthetic code design

2. What is a key advantage of using efficient data structures in terms of memory utilization?

- ☐ Increased memory consumption
- ☒ Reduced memory waste ✓
- ☐ No impact on memory

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Exam Administration Academics

(Credits - 03) Skill enhancement with Data structure algorithm - (C language)

33 lectures 9 sections 14 h 32 m 18 s

Quiz - 2

Total Score: 25 | Your Score: 25

1. Which of the following is an application of arrays?

- ☐ Implementing linked lists
- ☒ Representing a deck of cards ✓
- ☐ Storing a dynamic number of variables
- ☐ Defining conditional statements

2. Which keyword is commonly used to traverse an array in programming loops?

- ☐ next
- ☐ continue
- ☒ for ✓
- ☐ loop

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Exam Administration Academics

(Credits - 03) React Full stack (Web/App development Skills)

40 lectures 13 sections 9 h 43 m 32 s

Assignment 1

Total Score: 25 | Your Score: 24

What is React primarily used for?

- ☐ Server-side programming
- ☐ Styling web pages
- ☒ Building user interfaces ✓
- ☐ Database management

Which company developed React?

- ☐ Google
- ☒ Facebook ✓
- ☐ Microsoft
- ☐ Apple

React is a

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Assignment 2

Total Score: 25 | Your Score: 24

In React Bootstrap, which class is used to add a button at the bottom of a card?

- ☐ card-button
- ☐ btn-card
- ☒ card-footer ✓
- ☐ card-body

When is the useEffect Hook executed by default in a React component?

- ☐ Only after the initial render
- ☒ After every render, including initial render and updates ✓
- ☐ Only when a dependency changes
- ☐ Only before the component is removed from the DOM

In React, which tool or library is often used to handle HTTP requests and promises more easily when working with APIs?

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THANK YOU



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